

HAAS — How To Run This Application

For Supervisors, Examiners, and Students

Quick Start (30 seconds)

Option A: One-Click (Recommended)

1. Double-click `autorun.bat`
2. Wait for the setup to complete (~30 seconds first time)
3. Your browser will open automatically to `http://localhost:5000`
4. **Login:** Use any email (e.g., `test@uel.ac.uk`) and any password with 6+ characters (e.g., `123456`)
5. You're in! Explore the dashboard.

Option B: Manual (If Option A fails)

Open a Command Prompt/Terminal in this folder and run:

```
pip install flask fpdf2
python generate_data.py
python app.py
```

Then open your browser to `http://localhost:5000`

System Requirements

- **Python 3.10 or higher** (download from <https://www.python.org>)
 - **Any modern web browser** (Chrome, Edge, Firefox)
 - **No internet connection required** (the app runs entirely locally)
 - **Operating System:** Windows 10/11, macOS, or Linux
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What You'll See

1. Login Screen

- **This is a demo system** — there is no real authentication
- Enter any email address and any password (minimum 6 characters)
- Click "Sign In"

2. Dashboard (Command Centre)

- National statistics: total hospitals, available beds, CQC ratings
- Click the tabs to explore different sections

3. Key Demonstration Scenarios

Scenario A: Testing the CETR Gatekeeper (Admission Blocked)

1. Click the **"Risk Assessment"** tab
2. Enter: Age = **14**, Diagnosis = **Autism**, Mental Illness = **No**
3. Click "Check Admission"
4. **Expected Result:** The system BLOCKS the admission and shows community alternatives
5. **Why:** Under CETR guidelines, a child with autism and no mental illness cannot be admitted

Scenario B: Testing MCDA Hospital Matching (Admission Approved)

1. Click the **"Risk Assessment"** tab
2. Enter: Age = **35**, Diagnosis = **Schizophrenia**, Mental Illness = **Yes**
3. Click "Check Admission"
4. **Expected Result:** The system finds and ranks the top 5 hospitals by suitability score
5. **Why:** The patient has a treatable mental illness, so hospital admission is appropriate

Scenario C: Using the Interactive Tutorial

1. Click the  **Tutorial** button on the dashboard
2. Follow the step-by-step guided tour
3. It walks through both Scenario A and Scenario B automatically

4. Map View

- Click the **"Map"** tab
- See all 220 hospitals (blue markers) and 150 community services (green markers) on an interactive map
- Click any marker for details

5. Audit Trail

- Click the **"Audit"** tab
- Every decision made by the system is logged here with timestamps and reasoning
- This demonstrates clinical accountability

Troubleshooting

Problem	Solution
<code>python</code> not recognised	Install Python from python.org, tick "Add to PATH"
<code>pip</code> not found	Run <code>python -m pip install flask fpdf2</code> instead
Port 5000 in use	Edit <code>app.py</code> line 921: change <code>port=5000</code> to <code>port=5001</code>
Database error	Delete <code>data/hospitals.db</code> and run <code>python generate_data.py</code>
Browser doesn't open	Manually go to <code>http://localhost:5000</code>

File Structure

```
Software_Application/
├─ app.py                ← Main backend (922 lines, Flask)
├─ generate_data.py      ← Creates synthetic database
├─ evaluate_1000_scenarios.py ← Automated test suite
├─ requirements.txt      ← Python dependencies
├─ autorun.bat           ← One-click startup
├─ data/
│   └─ hospitals.db      ← SQLite database (auto-generated)
├─ templates/
│   └─ index_professional.html ← Complete frontend UI
└─ static/
    └─ (CSS/JS assets)
```